**NETWORK TRAFFIC ANALYSIS PROJECT**

## 1. Introduction

This project demonstrates capturing and analyzing network traffic using Wireshark, tcpdump, and Snort in Kali Linux to detect suspicious activity.

## 2. Objectives

- Capture live network traffic
- Analyze DNS, HTTP, and ICMP packets
- Detect suspicious patterns
- Simulate port scanning attack

## 3. Tools Used

- Kali Linux
- Wireshark
- tcpdump
- Snort
- Nmap

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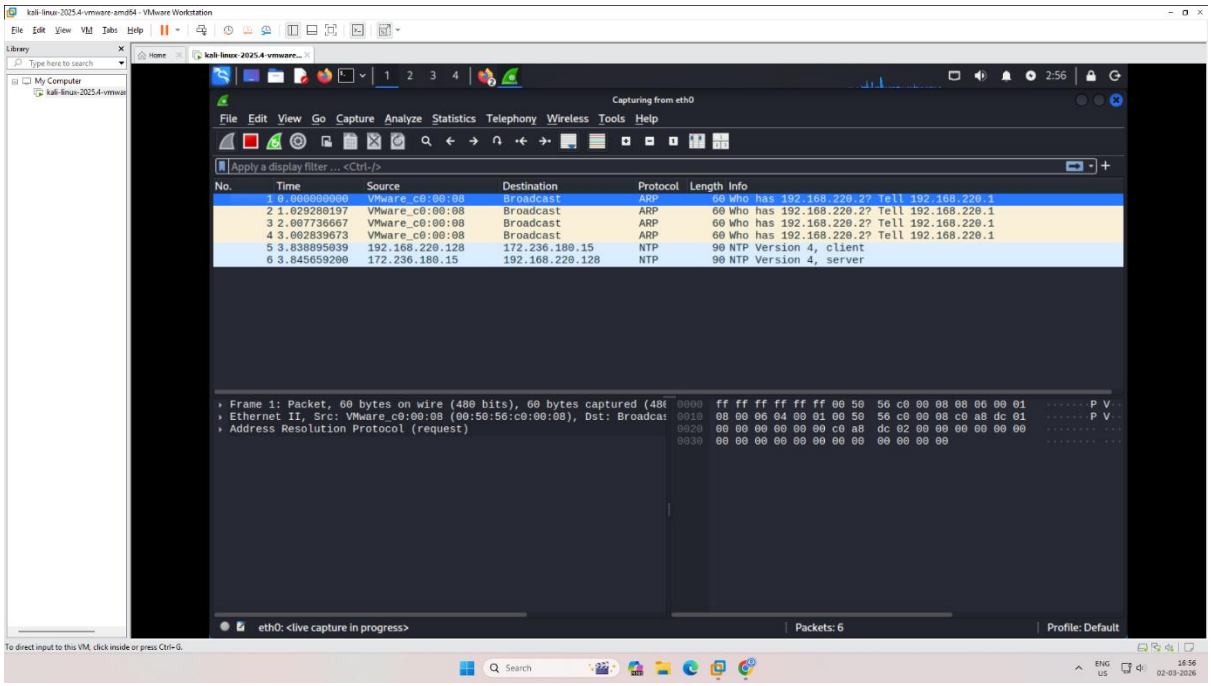
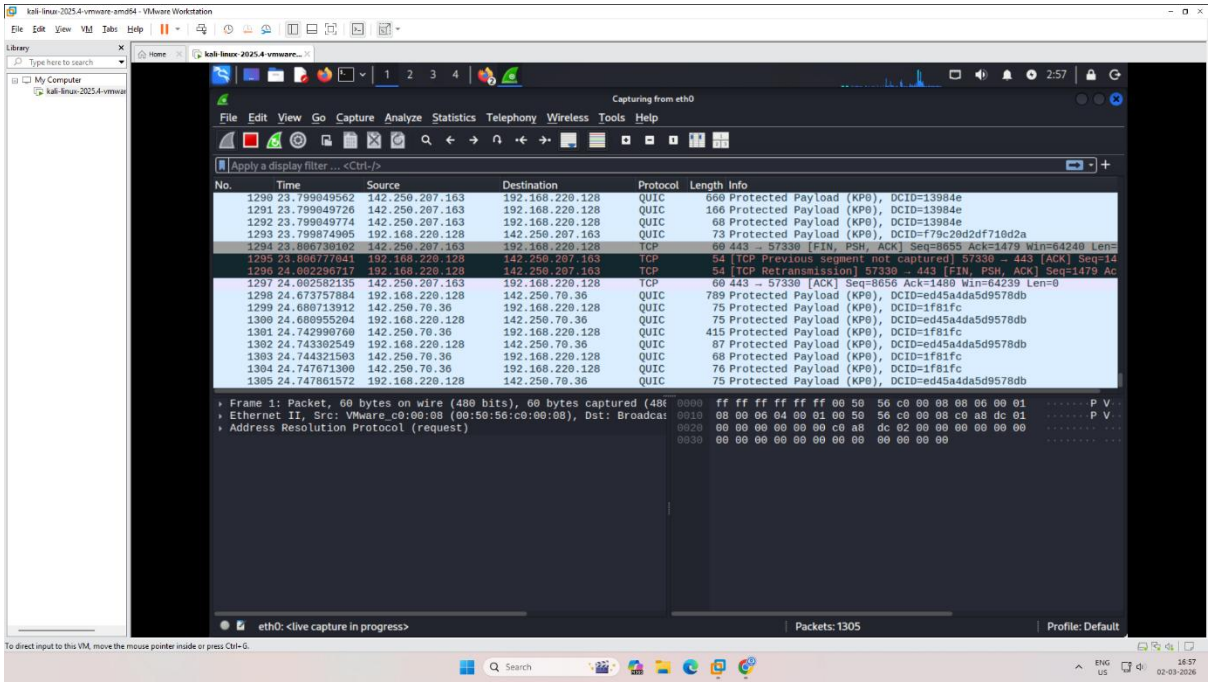
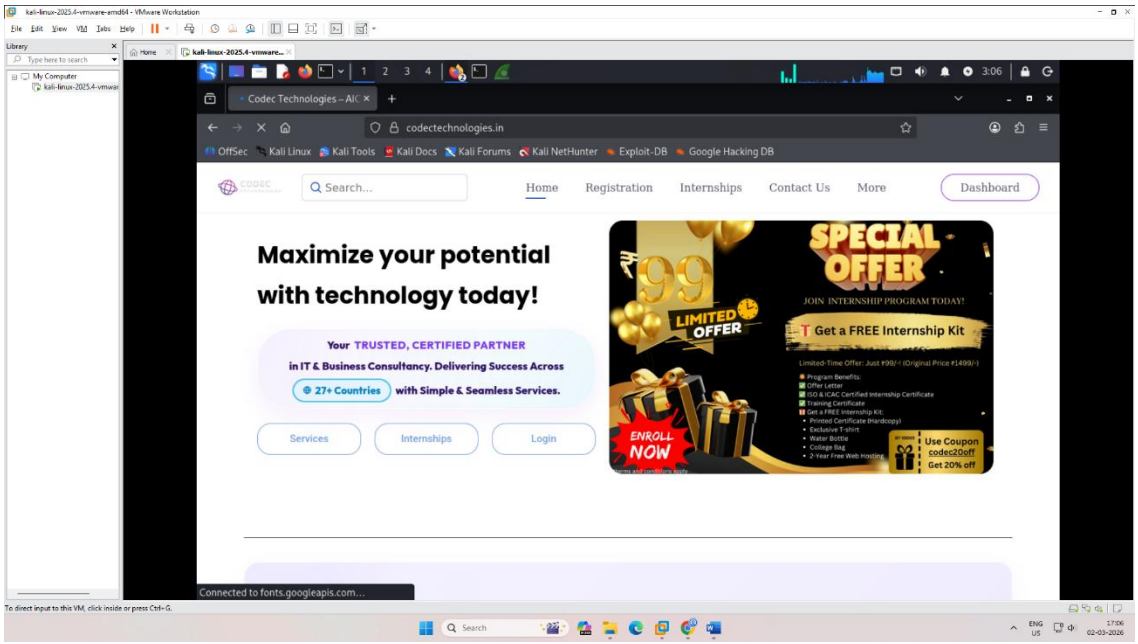
## 4. Methodology

### Step 1: Capturing Live Traffic

Wireshark was used to capture live packets from the active network interface.




kali-linux-2025.4-vmware-amd64 - VMware Workstation
Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1290 23.799049562 142.250.207.163 192.168.220.128 QUIC 660 Protected Payload (KP0), DCID=13984e
1291 23.799049726 142.250.207.163 192.168.220.128 QUIC 166 Protected Payload (KP0), DCID=13984e
1292 23.799049774 142.250.207.163 192.168.220.128 QUIC 68 Protected Payload (KP0), DCID=13984e
1293 23.799874905 192.168.220.128 142.250.207.163 QUIC 73 Protected Payload (KP0), DCID=f79c20d2df710d2a
1294 23.806730102 142.250.207.163 192.168.220.128 TCP 60 443 → 57330 [FIN, PSH, ACK] Seq=8655 Ack=1479 Win=64240 Len=
1295 23.806777041 192.168.220.128 142.250.207.163 TCP 54 [TCP Previous segment not captured] 57330 → 443 [ACK] Seq=14
1296 24.002296717 192.168.220.128 142.250.207.163 TCP 54 [TCP Retransmission] 57330 → 443 [FIN, PSH, ACK] Seq=1479 Ac
1297 24.002582135 142.250.207.163 192.168.220.128 TCP 60 443 → 57330 [ACK] Seq=8656 Ack=1480 Win=64239 Len=0
1298 24.673757884 192.168.220.128 142.250.70.36 QUIC 789 Protected Payload (KP0), DCID=ed45a4da5d9578db
1299 24.680713912 142.250.70.36 192.168.220.128 QUIC 75 Protected Payload (KP0), DCID=1f81fc
1300 24.680955204 192.168.220.128 142.250.70.36 QUIC 75 Protected Payload (KP0), DCID=ed45a4da5d9578db
1301 24.742990760 142.250.70.36 192.168.220.128 QUIC 415 Protected Payload (KP0), DCID=1f81fc
1302 24.743302549 192.168.220.128 142.250.70.36 QUIC 87 Protected Payload (KP0), DCID=ed45a4da5d9578db
1303 24.744321503 142.250.70.36 192.168.220.128 QUIC 68 Protected Payload (KP0), DCID=1f81fc
1304 24.747671300 142.250.70.36 192.168.220.128 QUIC 76 Protected Payload (KP0), DCID=1f81fc
1305 24.747861572 192.168.220.128 142.250.70.36 QUIC 75 Protected Payload (KP0), DCID=ed45a4da5d9578db
Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (48
Ethernet II, Src: VMware_c0:00:08 (00:50:56:c0:00:08), Dst: Broadca
Address Resolution Protocol (request)
eth0: <live capture in progress>
Packets: 1305
Profile: Default


kali-linux-2025.4-vmware-amd64 - VMware Workstation
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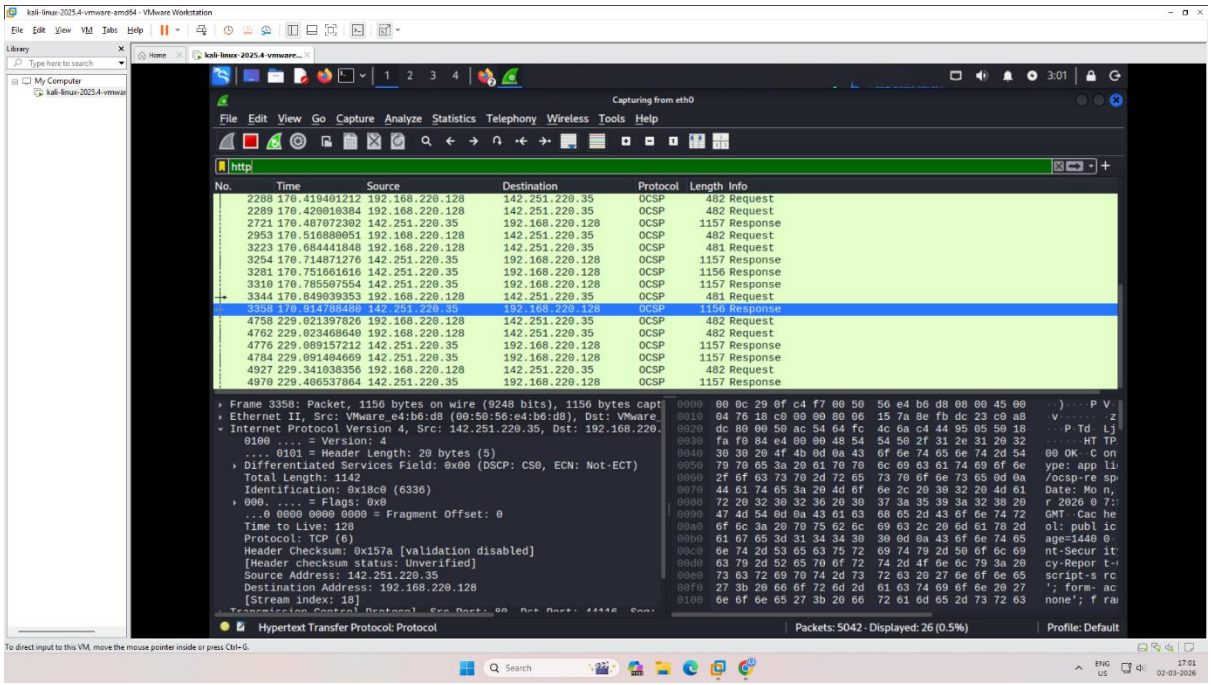
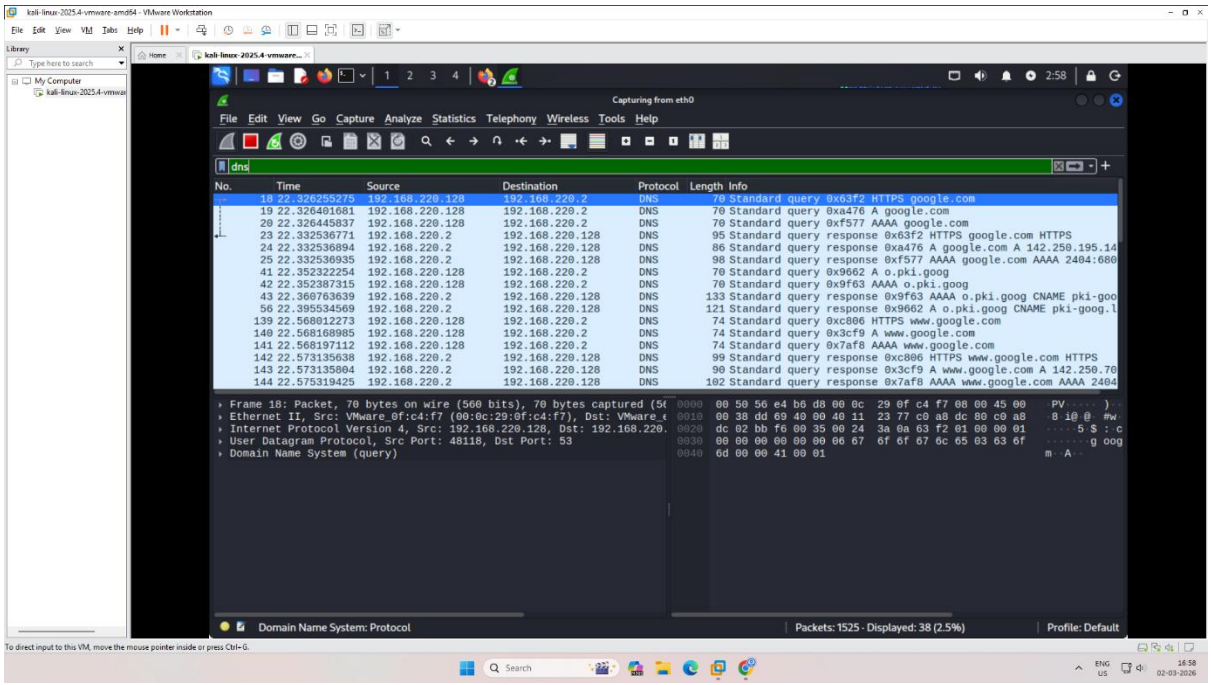

kali-linux-2025.4-vmware-amd64 - VMware Workstation
Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
2288 170.419401212 192.168.220.128 142.251.220.35 OCSP 482 Request
2289 170.420010384 192.168.220.128 142.251.220.35 OCSP 482 Request
2721 170.487072302 142.251.220.35 192.168.220.128 OCSP 1157 Response
2953 170.516880051 192.168.220.128 142.251.220.35 OCSP 482 Request
3223 170.684441848 192.168.220.128 142.251.220.35 OCSP 481 Request
3254 170.714871276 142.251.220.35 192.168.220.128 OCSP 1157 Response
3281 170.751661616 142.251.220.35 192.168.220.128 OCSP 1156 Response
3310 170.785507554 142.251.220.35 192.168.220.128 OCSP 1157 Response
3344 170.849039353 192.168.220.128 142.251.220.35 OCSP 481 Request
3358 170.914788480 142.251.220.35 192.168.220.128 OCSP 1156 Response
4758 229.021397826 192.168.220.128 142.251.220.35 OCSP 482 Request
4762 229.023468640 192.168.220.128 142.251.220.35 OCSP 482 Request
4776 229.089157212 142.251.220.35 192.168.220.128 OCSP 1157 Response
4784 229.091404669 142.251.220.35 192.168.220.128 OCSP 1157 Response
4927 229.341038356 192.168.220.128 142.251.220.35 OCSP 482 Request
4970 229.406537864 142.251.220.35 192.168.220.128 OCSP 1157 Response
Frame 3358: Packet, 1156 bytes on wire (9248 bits), 1156 bytes capt
Ethernet II, Src: VMware_e4:b6:d8 (00:50:56:e4:b6:d8), Dst: VMware
Internet Protocol Version 4, Src: 142.251.220.35, Dst: 192.168.220.
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1142
Identification: 0x18c0 (6336)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x157a [validation disabled]
[Header checksum status: Unverified]
Source Address: 142.251.220.35
Destination Address: 192.168.220.128
[Stream index: 18]
Hypertext Transfer Protocol: Protocol
Packets: 5042 · Displayed: 26 (0.5%)
Profile: Default

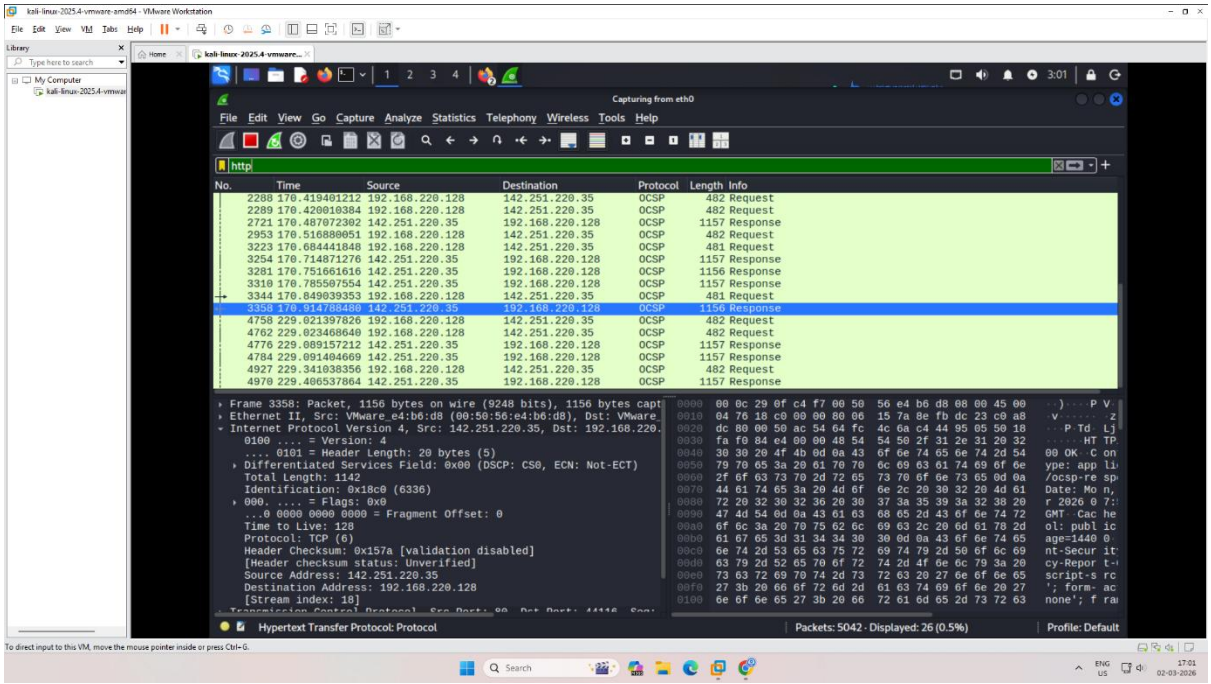
### Step 3: DNS Traffic Analysis

DNS packets were filtered using:




### Step 4: HTTP Traffic Analysis

HTTP traffic was filtered using:




### 5. Observations

- DNS queries were observed for domain resolution.
- HTTP packets contained plaintext requests.
- Port scan generated multiple SYN packets.
- Snort detected unusual traffic patterns.

- # 6. Conclusion
- Network traffic analysis helps detect suspicious activities such as port scanning and abnormal packet flows. Tools like Wireshark and Snort are effective for monitoring and intrusion detection.